Exploratory Data Analysis

Titanic Passenger Dataset | Internship Task 5

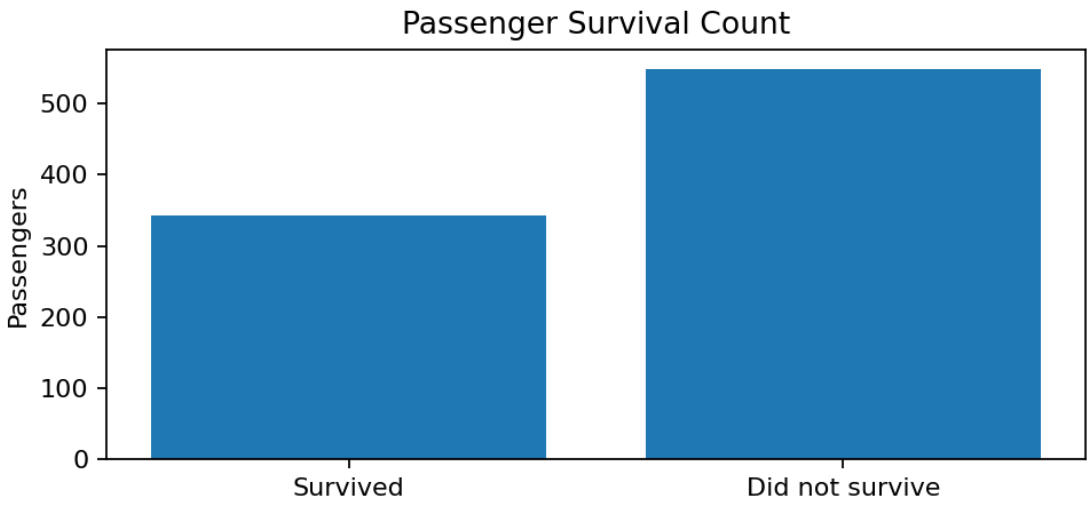
This report summarizes an exploratory data analysis of the Titanic passenger dataset. The analysis uses Python with Pandas, Matplotlib, and Seaborn to inspect the data, summarize key variables, visualize survival patterns, and identify limitations.

1. Dataset Overview

Measure	Result
Dataset size	891 rows × 12 columns
Passengers who survived	342 (38.4%)
Passengers who did not survive	549 (61.6%)
Missing age values	177
Missing cabin values	687
Missing embarkation values	2

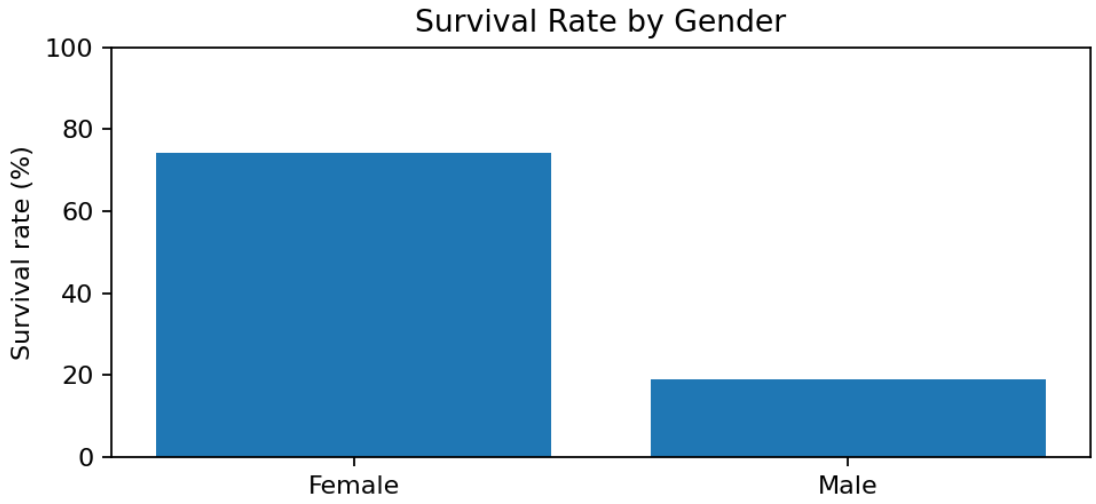
The dataset contains 891 passenger records and 12 columns. Initial checks included dataset shape, data types, summary statistics, value counts, and missing-value counts.

2. Overall Survival



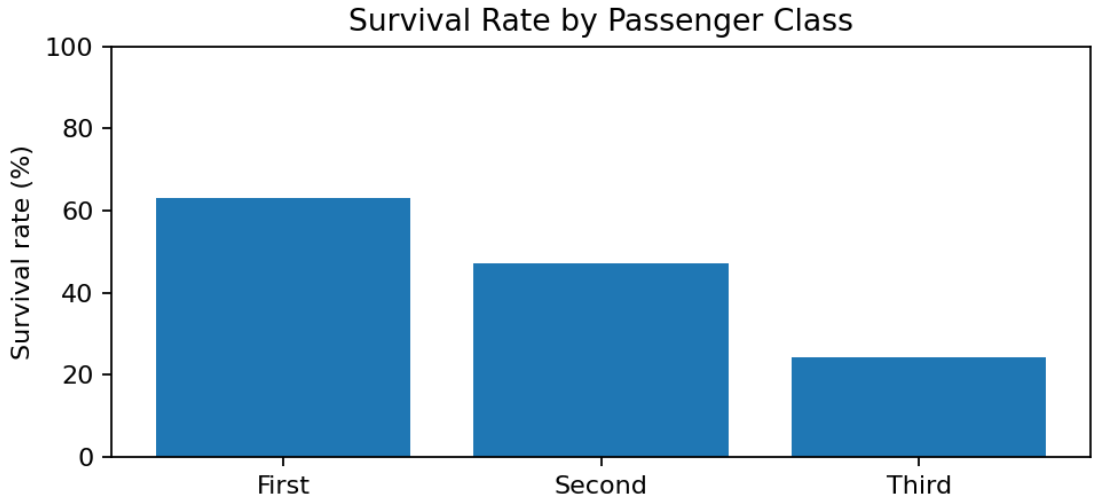
342 passengers survived and 549 did not. The survival rate was 38.4%, while 61.6% did not survive. Non-survivors were the larger group.

3. Survival by Gender



The observed survival rate was 74.2% for female passengers and 18.9% for male passengers. This is a strong difference in the dataset, but descriptive analysis alone cannot establish cause and effect.

4. Survival by Passenger Class



Observed survival rates were 63.0% for first class, 47.3% for second class, and 24.2% for third class. First-class passengers had the highest observed survival rate, and third-class passengers had the lowest.

5. Age and Fare Distribution

Age was missing for 177 passengers. Among available age values, the mean was 29.7 years and the median was 28 years. The age distribution is concentrated around younger and middle-aged adults. In the notebook's age-group analysis, children had the highest observed survival rate (about 58.0%), while seniors had the lowest (about 22.7%); passengers with missing ages were excluded.

Fare had a mean of 32.20 and a median of 14.45. The mean is much higher than the median, which is consistent with a right-skewed distribution: most fares were relatively low, while a smaller number were much higher. High fare values should be investigated rather than automatically removed.

6. Correlation and Pairwise Relationships

Variable pair	Correlation
Passenger class and fare	-0.55

Variable pair	Correlation
Survival and passenger class	-0.34
Survival and fare	+0.26
Siblings/spouses and parents/children aboard	+0.41

The correlation analysis showed a moderate negative relationship between passenger class and fare (about -0.55), a negative relationship between survival and passenger class (about -0.34), and a positive relationship between survival and fare (about +0.26). Siblings/spouses aboard and parents/children aboard had a positive correlation (about +0.41). The class variable is coded numerically, so its correlations require cautious interpretation. Correlation does not prove causation. The pairplot complements the heatmap by showing distributions and pairwise patterns for selected numerical variables.

7. Data Quality and Limitations

Missing values include 177 in Age, 687 in Cabin, and 2 in Embarked. The notebook also flagged 107 exact duplicate rows. These were not automatically removed, because identical rows do not necessarily mean they represent the same passenger. The available data does not provide enough evidence to confidently decide that every flagged row should be deleted.

8. Conclusion

The EDA found that survival outcomes differed substantially by gender and passenger class. Female passengers and first-class passengers had higher observed survival rates in this dataset. The age-group analysis showed a higher observed survival rate among children than seniors. Fare values were strongly right-skewed, and missing age and cabin information limit some comparisons. Summary statistics, visualizations, and correlation analysis helped identify patterns, but the findings are descriptive and should not be interpreted as proof of causation.

Tools: Python, Pandas, Matplotlib, Seaborn. Dataset: Titanic passenger dataset. This report is intended to accompany the Task 5 Jupyter Notebook.